

Effects of Wearing Knee Supports on the Detection Threshold and Applicability of Redirected Walking

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Figure 1: (a) Virtual environment: Participants walk toward a barrel. (b) Real environment: Participants walk in arc shape. (c) Band-type supporters used in Band condition (d) Soft-type supporters used in Soft condition

ABSTRACT

Redirected walking (RDW) is a locomotion technique that compresses expansive virtual reality (VR) spaces into smaller physical spaces by manipulating a user's position and orientation in the real world. Previous studies have shown that wearing a knee-tightening device can expand the detection threshold (DT) in RDW. However, issues such as discomfort and walking difficulty with such devices remain unresolved. In this study, we conducted a preliminary experiment to examine whether the DT could be similarly expanded by wearing more comfortable knee supports. Additionally, our main experiment investigated the effect of comfortable supports on RDW applicability. The results indicated that while the impact of knee supports on DT could not be confirmed, a significant influence on RDW applicability was observed.

Index Terms: Virtual reality, Redirected walking, Knee support

1 INTRODUCTION

As virtual reality (VR) has become increasingly accessible, there has been a growing demand in fields such as training and simulation for technology that enables walking in VR spaces with a sense of realism akin to real-world spaces. The method of "real-walking," which considers a user's actual location and posture in the real space for determining avatar and camera coordinates in a VR space, is considered to provide the most natural walking experience [14]. However, this approach requires a real space that is as large as the target VR space, which is not always feasible. To address this challenge, a technique called redirected walking (RDW) has been proposed [8]. RDW manipulates the spatial correspon-

dence between real and VR spaces, allowing free movement within a VR space larger than the corresponding physical space.

Despite its advantages in mitigating spatial constraints, RDW still faces challenges, particularly when users perceive manipulation. Awareness of discrepancies between the position and posture of their real body, sensed through proprioception, and their virtual position and posture, perceived visually, can increase simulator sickness and cognitive load [3]. Consequently, various RDW techniques have been explored to determine the differential threshold (DT), the minimum manipulation users can perceive.

For example, it has been established that when users walk in real space along an arc but perceive the arc as a straight line in VR space (curvature manipulation), they do not detect the manipulation if the radius of the arc is greater than 11.0 m, feeling as if they are actually walking a straight line [9]. This suggests that a real space of at least 11 meters in radius is required for a natural walking experience in an expansive VR environment.

Recent studies have aimed not only to expand DT but also to investigate the applicability of manipulations users perceive but find acceptable [9, 11] to achieve more efficient RDW. For example, it has been suggested that wearing a specific device on the knee during RDW can expand DT, making manipulations imperceptible at radii greater than 6.8 m [4]. However, existing knee-worn devices are inflexible, leading to discomfort and difficulty walking. In this study, we investigated whether using a knee support that is more comfortable but still applies pressure to the knee could also expand DT. In the preliminary experiment, we investigated the impact of knee supports on the DT of RDW. Based on these findings, the main experiment focused on examining the effect of knee supports on the applicability of RDW.

2 RELATED WORK

2.1 Redirected Walking

Walking is a core activity in both real environments (REs) and virtual environments (VEs). Numerous strategies have been developed to render the walking experience in VEs in a manner that is both authentic and immersive. A particularly effective approach is real walking [14], where a user's physical location and orientation in an RE are mirrored in the corresponding VE. This approach offers a

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highly realistic walking experience but requires that the size of the RE is greater than that of the VE.

To overcome this limitation, the concept of RDW was introduced [8]. RDW condenses an expansive VE to fit within a confined RE by adjusting real-world movements using various gains and reflecting the altered movements in the VE. These gains can involve translation, rotation, curvature, or bending, modifying different aspects of movement [13, 6].

Numerous RDW controllers have been devised to adapt diverse VEs to REs [15, 7, 1]. Among these, curvature gain is noted for its effective space compression in RDW.

2.1.1 Curvature Gain

The concept of gain in RDW techniques, including curvature gain, was first outlined by Steinicke et al. [13]. In this approach, although users perceive themselves as walking straight in the VE, their actual path in the RE is curved. The curvature gain g_C quantifies the arc curvature in the real space. The curvature gain can be defined in various ways. We adopt the definition $g_C := \frac{1}{r_{real}}$, where r_{real} denotes the radius of the guiding circle in the RE. Therefore, the curvature gain is expressed in units of m^{-1} . Curvature gain has proven to be an effective RDW strategy for squeezing vast VEs into limited REs and has been studied extensively.

2.1.2 Detection Threshold

The DT is the minimum intensity at which a user begins to perceive RDW manipulation. Manipulations exceeding this threshold can be recognized by users. Typically, the DT in RDW is assessed using a two-alternative forced choice (2AFC) task. Steinicke et al. [12] pioneered this area and conducted experiments to identify DTs for translation, rotation, and curvature gains through 2AFC tasks. They reported a DT at a curvature gain of $0.045 m^{-1}$. Subsequent studies have reported varying DTs for curvature gain, including $0.067 m^{-1}$ [2], $0.086 m^{-1}$ [5], and $5.2^\circ/m$ ($g_C = 0.091 m^{-1}$) [9].

2.1.3 Applicability

In recent years, alongside the psychophysical metrics of DT, research has increasingly focused on the applicability of RDW as a practical metric. While DT measures whether a user perceives manipulation, applicability evaluates whether a user finds the manipulation induced by RDW acceptable, regardless of their perception.

The concept of applicability in RDW was introduced by Schmitz et al. and Rietzler et al., who investigated it using different methodologies [11, 9]. Schmitz et al. identified the maximum feasible gain by continuously varying the RDW gain and having users report when they found it inapplicable. In contrast, Rietzler et al. assessed the applicability of various gains by exposing users to RDW at different gain levels and collecting their responses through questionnaires after each trial.

In our study, we chose to follow the approach of Rietzler et al., which we believe allows for a more quantitative evaluation, and conducted our experiments accordingly.

2.2 Effect of Knee-tightening Devices on the DT

In experiments conducted by Fukui et al., it was shown that the DT for curvature gain manipulation increased when a specific knee-tightening device was used [4]. They reported that the DT increased to $0.148 m^{-1}$ during RDW with curvature manipulation while wearing the device, as shown in Figure 2. While the device, which applies significant tightening to the knee, presents issues like discomfort and walking difficulty, its ability to enhance the DT of RDW by targeting the knee area is noteworthy.

To explore this concept further, we conducted a preliminary experiment using knee supports as comfortable tightening devices to examine whether they would result in an expansion of the DT of



Figure 2: Devices used in the study by Fukui et al. [4]



Figure 3: Virtual environment (left) and real environment (right) for the preliminary experiment

RDW. Based on the results of this preliminary experiment, we investigated the effects of knee supports on the applicability of RDW in our main experiment.

3 USER STUDY: PRELIMINARY EXPERIMENT

In a preliminary experiment, we investigated the effects of wearing knee supports on the DT for RDW using curvature manipulation. Two types of knee supports were prepared and their influence on the DT was measured using a 2AFC task. Assessments were conducted in scenarios in which participants wore each type of knee support, as well as in scenarios without any knee support.

3.1 Knee Supports

In our preliminary experiment, we evaluated two support types shown in Fig. 1 (c) and (d), referred to as the band and soft types, respectively. Each type came in three sizes—small (S), medium (M), and large (L)—with the appropriate size selected based on each participant's knee circumference. Hereafter, the band-type support is referred to as the band condition, the soft-type support as the soft condition, and the absence of support as the control condition.

3.2 Experimental Environment

Our experiment was conducted in a laboratory measuring $5, m \times 5, m$ (see Figure 3 (right)). The participants wore an HTC Vive Pro Head-Mounted Display (HMD) with a resolution of $1,440 \times 1,600$ pixels per eye, pixel density of 615 PPI, field of view of approximately 110° , and refresh rate of 90 Hz. An HTC Vive controller was used to provide responses to questionnaires.

In each trial, the VR environment was rendered using Unity 2019.4 and simulated a container yard (see Figure 3 (left)). This setting was chosen for its ample space and multiple objects serving as landmarks. Drum barrel objects were placed as landmarks for participants to aim at while walking.

The experimental environment and HMD used in our experiment were identical to those used in a previous study by Fukui et al. [4].

Each trial included four potential starting points. The point closest to the participant that ensured that the walking path remained within the experimental area was selected as the starting point for the next trial. This approach ensured that participants were unable to ascertain their actual positions and orientations in real space.

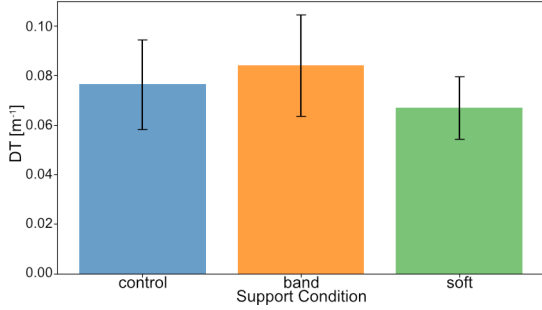


Figure 4: Average DT values (from left to right: control, band, and soft conditions). No significant main effect of support condition was confirmed.

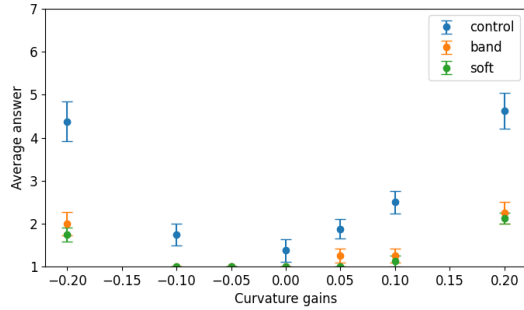


Figure 5: Example responses from a participant who experienced ambiguity in curvature perception (error bars indicate standard error), where the values associated with the band and soft conditions are smaller than those associated with the control condition

This also made it difficult for participants to estimate the magnitude of the gain applied in the previous trial and the curvature gain to be applied in the next trial. The subsequent trial commenced when the participant stood at the starting point facing straight ahead.

3.3 Conditions

For our experiment, we recruited the same participants that participated in the study by Fukui et al. [4]. The participants were five healthy individuals aged between 22 and 23 years (one female and four males; age: $M = 22.6$, $SD = 0.55$; body height: $M = 171.0$ cm, $SD = 6.0$ cm). None of the participants had impairments related to walking, vision, or hearing. All participants reported being right-footed, which was determined according to the foot that they would typically use to kick a ball.

The experiment included three within-participant conditions, namely the band, soft, and control conditions. For the curvature gain, seven levels were considered: $(0, \pm 0.05, \pm 0.1, \text{ and } \pm 0.2)$. Each level was trialed eight times, resulting in a total of 168 trials. The data for the control condition were taken from the paper by Fukui et al. [4], whereas new data were collected for the band and soft conditions in our experiment.

Data collection for the band and soft conditions was conducted using a block design in which all gain conditions under each support condition were completed before moving on to the gain conditions of the other support condition. The order of the soft and band conditions, and sequence of gain levels within each support condition were selected randomly. Additionally, the participants were not informed of the experimental purpose beforehand.

3.4 Procedure

All participants received a detailed explanation of the experimental procedure and provided informed consent. They also completed a



Figure 6: Virtual environment (left) and real environment (right) for the main experiment

questionnaire on basic information such as age, height, and dominant foot. Knee circumference measurements were taken to determine the appropriate knee support size, and participants selected shoes from sizes 23 to 27 cm.

The participants completed two practice trials wearing the HMD and headphones to fully understand the procedure. Subsequently, they proceeded to the main trials, wearing noise-cancelling headphones and an HMD. Each trial consisted of five stages. First, participants stood on a black arrow indicating the starting point and direction for the next trial. Next, the environment transitioned to the virtual container yard, and the target drum barrel appeared in their line of sight. Then, participants walked approximately 5 m toward the target as the headphones played white noise and a 100 BPM tempo for pacing, and the curvature gain was presented. Upon reaching the target, a post-trial questionnaire was displayed, and participants responded using the controller. Finally, the VE and target drum barrel were hidden from view.

The post-trial questionnaire asked two questions. The first question was, “Did you feel that your walking path was bent to the right or left?” with participants choosing either “Left” or “Right” (a 2AFC task). The second question was, “Did you feel that your walking path was bent?” Participants responded on a seven-point Likert scale, where one indicated no perception of bending and seven indicated strong bending.

After completing all trials, participants removed all equipment, including the knee supports, and responded to an oral interview.

3.5 Results and Discussion

3.5.1 DT

Figure 4 presents the average DTs obtained for each participant in the experiment. To calculate the DT, *psignifit* 4 [10] was used. The data for each participant were fitted using a cumulative Gaussian function. An analysis of variance (ANOVA) with one factor corresponding to the support condition (band, soft, or control) as the independent variable and DT as the dependent variable revealed no statistically significant main effects ($p = 0.79$, $F = 3.89$).

3.5.2 Curvature Perception

With regard to curvature perception, no significant differences were found based on the presence or absence of supports. To analyze participant responses, a two-way ANOVA with repeated measures was performed considering the support conditions (control, band, soft) and gains $(0, \pm 0.05, \pm 0.1, \text{ and } \pm 0.2)$ as independent variables, and the questionnaire responses as the dependent variable.

As shown in Figure 5, differences in the perception of walking curvature were observed between the support conditions for a participant. In the post-trial interviews, the participant also mentioned that the sense of walking curvature was weaker under the support-wearing conditions. However, no differences in DT values were observed between the support conditions for the participant.

This suggests that while the perception of walking direction can be maintained, wearing supports may suppress the perception of the

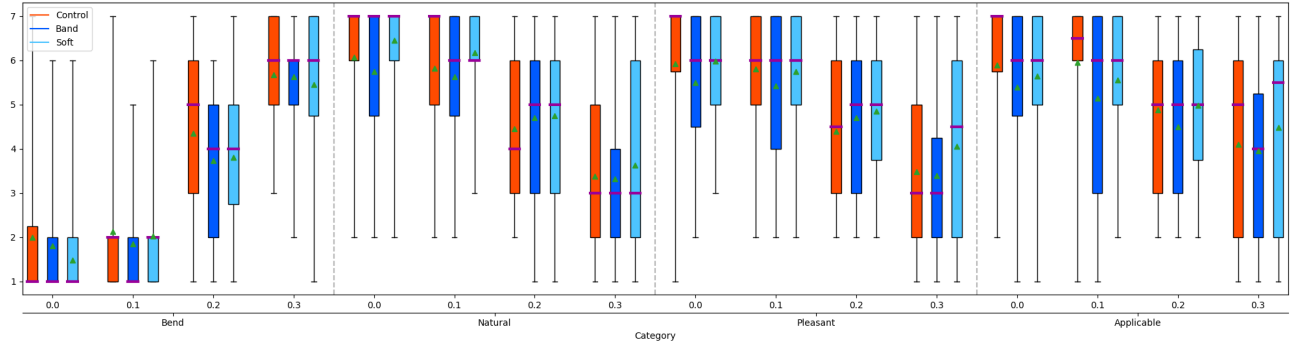


Figure 7: Box plot representation of responses to the questionnaire across various gains and support conditions. The median is indicated by the purple line, whereas the mean is represented by the green triangle

curvature of a walking path.

4 USER STUDY: MAIN EXPERIMENT

The results of our preliminary experiment suggest that knee supports may blur the perception of walking curvature in RDW, indicating the potential for enhancing the applicability of RDW. Consequently, in our main experiment, we investigated the effect of wearing knee supports on the applicability of RDW.

The experimental environment was similar to that used in the preliminary experiment. The primary difference was that different settings were used for the RE and VE (see Figs. 6).

4.1 Conditions

Ten healthy participants aged 20 to 24 years (two women and eight men; age: $M = 22.2$, $SD = 1.4$; height: $M = 170.9$ cm, $SD = 5.0$ cm) participated in the experiment. All participants were free of gait, visual, or hearing impairments. All participants indicated that they had a dominant right foot.

A total of 48 trials were conducted for each participant. Three support conditions (control, band, soft) and four curvature gain conditions (0.0, 0.1, 0.2, 0.3) were tested with four repetitions. The order of conditions experienced by participants was randomized.

4.2 Procedure

The participants followed the same procedure as in the preliminary experiment. However, a notable difference was that after each trial, participants responded to a set of questions designed to evaluate their experience. These questions included: (1) “How bent was your walking path in the real environment?” (Bent), (2) “Walking in this manner through a virtual world is natural.” (Natural), (3) “Walking in this manner through a virtual world is pleasant.” (Pleasant), and (4) “I could imagine using this walking technique to move inside virtual worlds.” (Applicable).

Participants answered these questions on a scale from one to seven, where one indicated “not at all” for the first question and “totally disagree” for the others, while seven indicated “very much” for the first question and “totally agree” for the others. These questions were identical to those used in the study by Rietzler et al. [9].

After completing all trials, participants were interviewed about their experiences and completed oral questionnaires.

4.3 Results and Discussion

Figure 7 presents a box plot detailing the participant responses to the questionnaire under each experimental condition.

To analyze these responses, a two-way ANOVA with repeated measures was conducted, considering the support conditions (control, band, soft) and gains (0.0, 0.1, 0.2, 0.3) as independent variables and the questionnaire responses as the dependent variable.

This analysis revealed a statistically significant main effect of support conditions on the “Applicable” responses ($p < .05$). Post-hoc analysis with the Bonferroni-adjusted Wilcoxon signed-rank test showed significant differences between the band and soft conditions ($p < .01$) and the band and control conditions ($p < .01$).

Interview feedback indicated a preference for the control and soft conditions over the band condition regarding ease of walking. Notably, several participants mentioned feeling unfamiliar and less stable with the band-type support, while the soft-type support appeared to provide better knee support, improving ease and steadiness when walking with uneven steps. These observations suggest that increased leg stability may reduce the conscious effort needed for walking, potentially enhancing the applicability of RDW.

However, these findings differ from the implications of the preliminary experiment. One possible reason for this discrepancy may be that many participants were not accustomed to wearing a band-type support. Several participants also mentioned the instability of walking under the band condition during the post-experiment interviews. For those unaccustomed to wearing band-type supports, it is conceivable that they actually required more attention while walking, potentially diminishing the applicability of RDW.

5 CONCLUSIONS AND FUTURE WORK

In this study, we first investigated the impact of knee supports on the DT in a preliminary experiment and, based on the results, we examined the influence of knee supports on the applicability of RDW in our main experiment.

The results of the preliminary experiment did not reveal an impact of different support conditions on the DT; however, they suggested an effect of rendering the perception of bending ambiguous, which, in turn, implied a potential enhancement in RDW applicability. In the main experiment, although the outcomes differed from our preliminary expectations, it was demonstrated that wearing knee supports affected the applicability of RDW.

A promising area of investigation is identifying specific aspects of supports that enhance applicability and exploring designs to maximize this effect. Additionally, since applying various knee pressure values improved RDW applicability, applying similar pressure to other joints, such as the elbows, might enhance redirection for other body parts (e.g., hand redirection and haptic retargeting). Therefore, further studies on other joints are warranted.

ACKNOWLEDGMENTS

This work was partially supported by a JSPS Grant-in-Aid for Scientific Research (B) (JP19H04149, JP22H03628), Grant-in-Aid for Early-Career Scientists (22K17929), JST ASPIRE (JPM-JAP2404).

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